

Multi-Agent System for Smart Appliance Scheduling: Proposed Methods and Experimental Setup

Authors
Department
University
City, Country
email@domain.com

Abstract—This paper presents a comprehensive multi-agent system for intelligent appliance scheduling in smart homes. The proposed framework integrates four specialized agents: electricity price forecasting, household power consumption forecasting, appliance profiling, and scheduling optimization. We detail the mathematical formulations, algorithms, and experimental setup for each component, demonstrating how deep learning and optimization techniques can be combined to minimize energy costs while respecting operational constraints.

I. INTRODUCTION

The proliferation of smart home technologies necessitates intelligent energy management systems that can optimize appliance scheduling based on dynamic electricity pricing and household consumption patterns. This paper presents a multi-agent architecture where each agent specializes in a specific aspect of the scheduling problem.

II. PROPOSED MULTI-AGENT ARCHITECTURE

A. System Overview

The proposed system consists of four interconnected agents:

- **Agent 1:** Electricity Price Forecasting
- **Agent 2:** Household Power Forecasting
- **Agent 3:** Appliance Profiling
- **Agent 4:** Scheduling Optimizer

Figure ?? illustrates the information flow between agents, where Agents 1-3 provide forecasts and profiles to Agent 4, which performs the final scheduling optimization.

III. AGENT 1: ELECTRICITY PRICE FORECASTING

A. Problem Formulation

Agent 1 predicts the electricity price $p_t \in \mathbb{R}^+$ (in \$/kWh) at hour t on day d , where $d \in [1, 365]$ and $t \in [1, 24]$.

B. Feature Engineering

To capture the cyclical nature of time, we employ sinusoidal transformations:

$$\mathbf{x}_{hour} = \begin{bmatrix} \sin\left(\frac{2\pi t}{24}\right) \\ \cos\left(\frac{2\pi t}{24}\right) \end{bmatrix} \quad (1)$$

$$\mathbf{x}_{day} = \begin{bmatrix} \sin\left(\frac{2\pi d}{365}\right) \\ \cos\left(\frac{2\pi d}{365}\right) \end{bmatrix} \quad (2)$$

The complete feature vector is:

$$\mathbf{x} = \left[\mathbf{x}_{hour}^T, \mathbf{x}_{day}^T, \frac{t}{24}, \frac{d}{365} \right]^T \in \mathbb{R}^6 \quad (3)$$

C. Neural Network Architecture

We employ a feed-forward neural network with the following architecture:

$$f(\mathbf{x}) = \mathbf{W}_4 \cdot \sigma(\mathbf{W}_3 \cdot \sigma(\mathbf{W}_2 \cdot \sigma(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2) + \mathbf{b}_3) + b_4 \quad (4)$$

where:

- Input layer: 6 neurons
- Hidden layer 1: 64 neurons
- Hidden layer 2: 32 neurons
- Hidden layer 3: 16 neurons
- Output layer: 1 neuron
- Activation function: $\sigma(z) = \max(0, z)$ (ReLU)

D. Training Procedure

The network is trained using the Adam optimizer to minimize the Mean Squared Error (MSE) loss:

$$\mathcal{L}_{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (5)$$

where y_i is the actual price and $\hat{y}_i$ is the predicted price.

E. Data Normalization

Both features and targets are standardized using:

$$z = \frac{x - \mu}{\sigma} \quad (6)$$

where μ and σ are the mean and standard deviation computed from the training set.

IV. AGENT 2: HOUSEHOLD POWER FORECASTING

A. Problem Formulation

Agent 2 predicts household power consumption P_t (in kW) at hour t on day d .

B. Enhanced Feature Engineering

Building upon Agent 1's features, we add monthly cyclical encoding:

$$m = \left\lceil \frac{d}{30.5} \right\rceil \quad (7)$$

$$\mathbf{x}_{month} = \begin{bmatrix} \sin\left(\frac{2\pi m}{12}\right) \\ \cos\left(\frac{2\pi m}{12}\right) \end{bmatrix} \quad (8)$$

The complete feature vector becomes:

$$\mathbf{x} = \left[\mathbf{x}_{hour}^T, \mathbf{x}_{day}^T, \mathbf{x}_{month}^T, \frac{t}{24}, \frac{d}{365} \right]^T \in \mathbb{R}^8 \quad (9)$$

C. Data Aggregation

Minute-level power consumption data is aggregated to hourly resolution:

$$P_{d,t} = \frac{1}{|\mathcal{T}_{d,t}|} \sum_{i \in \mathcal{T}_{d,t}} P_i \quad (10)$$

where $\mathcal{T}_{d,t}$ is the set of all minute-level readings within hour t of day d .

D. Neural Network Architecture

Similar to Agent 1, but adapted for 8 input features:

- Input layer: 8 neurons
- Hidden layers: $64 \rightarrow 32 \rightarrow 16$ neurons
- Output layer: 1 neuron
- Activation: ReLU
- Optimizer: Adam

V. AGENT 3: APPLIANCE PROFILING

A. Appliance Categorization

Appliances are classified into three categories:

$$C(a) = \begin{cases} \text{ESSENTIAL} & \text{if } a \in \mathcal{A}_E \\ \text{NECESSARY} & \text{if } a \in \mathcal{A}_N \\ \text{EXPENDABLE} & \text{if } a \in \mathcal{A}_X \end{cases} \quad (11)$$

where:

$$\mathcal{A}_E = \{\text{Fridge, Heater, AC, Lights}\} \quad (12)$$

$$\mathcal{A}_N = \{\text{Oven, Microwave, TV, Computer}\} \quad (13)$$

$$\mathcal{A}_X = \{\text{Washing Machine, Dishwasher}\} \quad (14)$$

B. Statistical Profile Construction

For each appliance a , we compute:

1) Energy Statistics:

$$E_{avg}(a) = \frac{1}{N_a} \sum_{i=1}^{N_a} E_i \quad (15)$$

$$E_{std}(a) = \sqrt{\frac{1}{N_a - 1} \sum_{i=1}^{N_a} (E_i - E_{avg}(a))^2} \quad (16)$$

2) Hourly Usage Frequency:

$$f_t(a) = \frac{|\{i : h_i = t, a_i = a\}|}{N_a} \quad (17)$$

where h_i is the hour of occurrence i and N_a is the total number of occurrences for appliance a .

3) Peak Hours Identification:

$$\mathcal{H}_{peak}(a) = \text{argtop-k}_{t \in [1, 24]} f_t(a) \quad (18)$$

where argtop-k returns the k hours with highest frequency.

4) *Power and Duration Estimation:* For schedulable appliances, we estimate:

$$\hat{P}(a) = \frac{E_{avg}(a)}{\hat{\tau}(a)} \quad (19)$$

where $\hat{\tau}(a)$ is the estimated duration:

$$\hat{\tau}(a) = \begin{cases} 2.0 & \text{if } a \in \{\text{Washing Machine, Dishwasher}\} \\ 0.5 & \text{if } a \in \{\text{Oven, Microwave}\} \\ 3.0 & \text{if } a \in \{\text{TV, Computer}\} \\ 1.0 & \text{otherwise} \end{cases} \quad (20)$$

VI. AGENT 4: SCHEDULING OPTIMIZER

A. Problem Formulation

Given:

- Set of appliances $\mathcal{A}$
- Price forecast p_t for $t \in [1, 24]$
- Power forecast P_t for $t \in [1, 24]$
- Appliance profiles $\{E(a), \hat{P}(a), \hat{\tau}(a)\}_{a \in \mathcal{A}}$
- Maximum power limit P_{max}

Objective: Minimize total cost while respecting power constraints.

B. Linear Programming Formulation

1) Decision Variables:

$$x_{a,t} \in \{0, 1\}, \quad \forall a \in \mathcal{A}, t \in [1, 24] \quad (21)$$

where $x_{a,t} = 1$ if appliance a starts at hour t .

2) Objective Function:

$$\min \sum_{a \in \mathcal{A}} \sum_{t=1}^{24} x_{a,t} \cdot E(a) \cdot p_t \quad (22)$$

3) Constraints: Single Execution Constraint:

$$\sum_{t=1}^{24} x_{a,t} = 1, \quad \forall a \in \mathcal{A} \quad (23)$$

Power Limit Constraint:

$$P_t + \sum_{a \in \mathcal{A}} \sum_{\substack{s=1 \\ s \leq t < s + \lceil \hat{\tau}(a) \rceil}}^{24} x_{a,s} \cdot \hat{P}(a) \leq P_{max}, \quad \forall t \in [1, 24] \quad (24)$$

This constraint ensures that at any hour t , the baseline power consumption P_t plus the power from all running appliances does not exceed P_{max} .

C. Reinforcement Learning Greedy Heuristic

Algorithm 1 RL Greedy Scheduling

```

1: Input:  $\mathcal{A}, \{p_t\}, \{P_t\}, P_{max}$ 
2: Sort hours:  $H_{sorted} = \text{argsort}_t(p_t)$ 
3: Initialize:  $S \leftarrow \emptyset, H_{used} \leftarrow \emptyset$ 
4: for each  $a \in \mathcal{A}$  do
5:    $cost_{best} \leftarrow \infty, t_{best} \leftarrow \text{null}$ 
6:   for each  $t \in H_{sorted}$  do
7:     if  $t \in H_{used}$  or  $t + \lceil \hat{\tau}(a) \rceil > 24$  then
8:       continue
9:     end if
10:    if  $P_t + \hat{P}(a) > P_{max}$  then
11:      continue
12:    end if
13:     $cost \leftarrow E(a) \cdot p_t$ 
14:    if  $cost < cost_{best}$  then
15:       $cost_{best} \leftarrow cost, t_{best} \leftarrow t$ 
16:    end if
17:  end for
18:  if  $t_{best} \neq \text{null}$  then
19:     $S \leftarrow S \cup \{(a, t_{best})\}$ 
20:     $H_{used} \leftarrow H_{used} \cup \{t_{best}, \dots, t_{best} + \lceil \hat{\tau}(a) \rceil - 1\}$ 
21:  end if
22: end for
23: Return:  $S$ 

```

1) Algorithm:

D. Baseline Schedule

For comparison, we define a baseline schedule based on peak usage hours:

$$t_{baseline}(a) = \mathcal{H}_{peak}(a)[0] \quad (25)$$

The baseline cost is:

$$C_{baseline} = \sum_{a \in \mathcal{A}} E(a) \cdot p_{t_{baseline}(a)} \quad (26)$$

E. Performance Metrics

We evaluate scheduling performance using:

1) Cost Reduction:

$$\Delta C = C_{baseline} - C_{optimized} \quad (27)$$

2) Savings Percentage:

$$\eta = \frac{\Delta C}{C_{baseline}} \times 100\% \quad (28)$$

3) Optimality Gap (LP vs. RL):

$$\gamma = \frac{C_{RL} - C_{LP}}{C_{LP}} \times 100\% \quad (29)$$

VII. EXPERIMENTAL SETUP

A. Datasets

1) Price Forecasting Dataset:

- **Source:** Historical electricity market data
- **Features:** Day (1-365), Hour (1-24)
- **Target:** Electricity Price (\$/MWh $\rightarrow$ \$/kWh)
- **Size:** 8,760 hourly records (1 year)
- **Split:** 70% train, 15% validation, 15% test

2) Power Forecasting Dataset:

- **Source:** Individual household electric power consumption
- **Resolution:** Minute-level aggregated to hourly
- **Features:** Day, Hour, Seasonal indicators
- **Measurements:** Global active power, voltage, intensity, sub-metering
- **Split:** 70% train, 15% validation, 15% test

3) Device Consumption Dataset:

- **Source:** Smart home appliance usage logs
- **Features:** Date, Time, Home ID, Appliance Type, Energy Consumption, Temperature, Season
- **Appliances:** 10 types across multiple homes
- **Records:** Multiple observations per appliance type

B. Training Configuration

TABLE I
NEURAL NETWORK TRAINING PARAMETERS

Parameter	Value
Optimizer	Adam
Learning rate	Default (0.001)
Max iterations	500
Hidden layers	[64, 32, 16]
Activation	ReLU
Batch size	Full batch
Random seed	42

1) Neural Network Hyperparameters:

2) Evaluation Metrics: For forecasting agents (1 and 2):

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} \quad (30)$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (31)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (32)$$

$$MAPE = \frac{100\%}{N} \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (33)$$

C. Scheduling Optimization Configuration

1) Test Scenarios:

- **Test Day:** Day 45 (mid-February)
- **Appliances:** Washing Machine, Dishwasher, Oven, Microwave, TV, Computer
- **Power Limit:** $P_{max} = 10.0$ kW
- **Methods:** LP (exact), RL Greedy (heuristic), Baseline (peak hours)

2) LP Solver Configuration:

- **Library:** PuLP (Python Linear Programming)
- **Solver:** CBC (COIN-OR Branch and Cut)
- **Variable type:** Binary
- **Problem type:** Mixed-Integer Linear Programming (MILP)

D. Implementation Details

1) Software Stack:

- **Language:** Python 3.8+
- **ML Framework:** scikit-learn 1.0+
- **Optimization:** PuLP 2.6+
- **Data Processing:** pandas 1.3+, NumPy 1.21+
- **Visualization:** matplotlib 3.4+, seaborn 0.11+

2) *Agent Integration:* The multi-agent system follows a sequential execution model:

1) Training Phase:

- Agent 1 trains on price data
- Agent 2 trains on power data
- Agent 3 builds profiles from device data
- All agents save models/profiles

2) Inference Phase:

- Agent 4 loads trained agents 1-3
- For target day d :
 - Query Agent 1 for $\{p_t\}_{t=1}^{24}$
 - Query Agent 2 for $\{P_t\}_{t=1}^{24}$
 - Query Agent 3 for appliance profiles
 - Execute LP and RL optimization
 - Compare against baseline

E. Performance Analysis

1) Quantitative Metrics:

- Forecasting accuracy (R^2 , MAE, RMSE)
- Scheduling cost reduction (\$/day)
- Savings percentage (%)
- Optimality gap (LP vs. RL)
- Annual savings projection (\$/year)

2) Qualitative Analysis:

- Appliance schedule visualization
- Price-schedule alignment
- Power constraint satisfaction
- Temporal distribution of scheduled appliances

F. Validation Strategy

1) Model Validation:

- Train-validation-test split (70-15-15)
- Cross-validation on forecasting models
- Residual analysis for prediction quality

2) Scheduling Validation:

- Constraint satisfaction verification
- Feasibility checking (all appliances scheduled)
- Solution optimality verification (LP solver status)
- Heuristic quality assessment (RL vs. LP gap)

VIII. EXPERIMENTAL WORKFLOW

The complete experimental pipeline executes as follows:

Algorithm 2 Multi-Agent System Execution

```
1: Phase 1: Agent Training
2: Load price data → Train Agent 1 → Save model
3: Load power data → Train Agent 2 → Save model
4: Load device data → Build Agent 3 profiles → Save profiles
5:
6: Phase 2: Integration Test
7: Load all trained agents
8: Verify agent functionality with test queries
9:
10: Phase 3: Scheduling Optimization
11: for each test scenario  $(d, \mathcal{A}, P_{max})$  do
12:    $\{p_t\} \leftarrow \text{Agent1.predict\_day}(d)$ 
13:    $\{P_t\} \leftarrow \text{Agent2.predict\_day}(d)$ 
14:   profiles  $\leftarrow \text{Agent3.get\_profiles}(\mathcal{A})$ 
15:    $S_{baseline} \leftarrow \text{baseline\_schedule}(\mathcal{A}, \text{profiles})$ 
16:    $S_{LP} \leftarrow \text{LP\_optimize}(d, \mathcal{A}, \{p_t\}, \{P_t\}, P_{max})$ 
17:    $S_{RL} \leftarrow \text{RL\_greedy}(d, \mathcal{A}, \{p_t\}, \{P_t\}, P_{max})$ 
18:   Compute costs and savings for all methods
19:   Generate visualizations and reports
20: end for
21:
22: Phase 4: Performance Analysis
23: Aggregate results across scenarios
24: Statistical significance testing
25: Sensitivity analysis on  $P_{max}$ 
```

IX. REPRODUCIBILITY

All experiments are designed for full reproducibility:

- Fixed random seeds (seed=42)
- Standardized train-test splits
- Documented hyperparameters
- Version-controlled code
- Saved model checkpoints
- Deterministic optimization solvers

X. CONCLUSION

This experimental setup provides a comprehensive framework for evaluating the proposed multi-agent appliance scheduling system. The combination of neural network forecasting, statistical profiling, and optimization techniques enables systematic analysis of energy cost reduction potential in smart homes.

ACKNOWLEDGMENT

The authors would like to thank the anonymous reviewers for their valuable comments and suggestions.

REFERENCES

- [1] Author, "Title," *Journal*, vol. X, no. Y, pp. Z, Year.